

Question 1: Scatterplot with Regression Diagnostics

```
library(ggplot2)

data(mtcars)

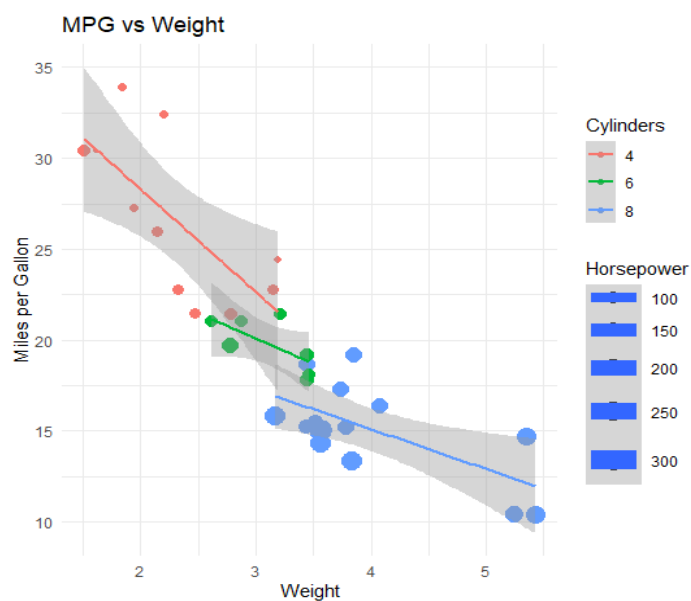
mtcars$cyl <- as.factor(mtcars$cyl)

ggplot(mtcars, aes(x = wt, y = mpg, color = cyl, size = hp)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "MPG vs Weight",
    x = "Weight",
    y = "Miles per Gallon",
    color = "Cylinders",
    size = "Horsepower"
  ) +
  theme_minimal()

correlation <- cor(mtcars$mpg, mtcars$wt, method = "pearson")

# Print correlation

print(correlation)
```

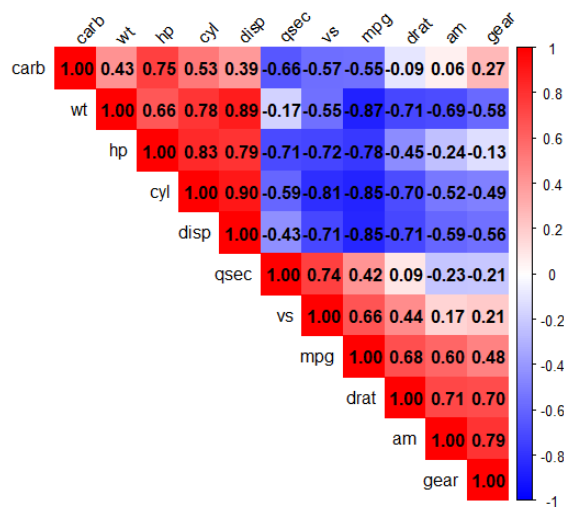


Question 2: Correlogram Construction

```

install.packages("corrplot")
library(corrplot)
data(mtcars)
cor_matrix <- cor(mtcars)
print(cor_matrix)
corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  order = "hclust",
  addCoef.col = "black",
  tl.col = "black",
  tl.srt = 45,
  col = colorRampPalette(c("blue", "white", "red"))(200)
)

```



Question 4: Bubble Chart for Multivariate Paired Data

```

install.packages("ggplot2")
install.packages("gapminder")
library(ggplot2)
library(gapminder)

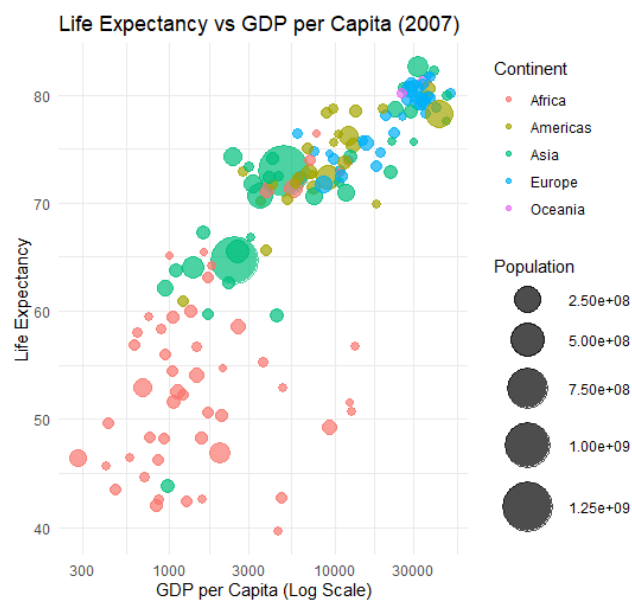
```

```

data2007 <- subset(gapminder, year == 2007)

ggplot(data2007, aes(
  x = gdpPercap,
  y = lifeExp,
  size = pop,
  color = continent
)) +
  geom_point(alpha = 0.7) +
  scale_x_log10() +
  scale_size_continuous(range = c(2, 15)) +
  labs(
    title = "Life Expectancy vs GDP per Capita (2007)",
    x = "GDP per Capita (Log Scale)",
    y = "Life Expectancy",
    size = "Population",
    color = "Continent"
  ) +
  theme_minimal()

```



Question 5: Hexbin Plot vs. Network Graph

```
install.packages("igraph")
```

```

library(igraph)
edges <- data.frame(
  from = c("A","A","A","B","B","C","C","D","D","E","E","F","F","G","H"),
  to   = c("B","C","D","C","E","D","F","E","G","F","H","G","H","H","A"),
  weight = c(5,3,4,6,2,7,3,5,4,6,2,5,8,3,4)
)
g <- graph_from_data_frame(
  edges,
  directed = FALSE
)
centrality <- degree(g)
print(centrality)
highest_node <- names(which.max(centrality))
cat("Node with highest centrality:", highest_node, "\n")
V(g)$size <- centrality * 6 + 10
E(g)$width <- E(g)$weight
plot(
  g,
  vertex.label = V(g)$name,
  vertex.color = "lightblue",
  edge.color = "gray",
  main = "Weighted Network Graph"
)

```

Weighted Network Graph

